S04 — Statistical learning of colour triplets: report

Question. After watching a stream of coloured nonsense shapes in which colours recur in fixed temporal triplets, do people judge colour-only triplets that occurred in the stream as more familiar than foil triplets, above chance (50%)?

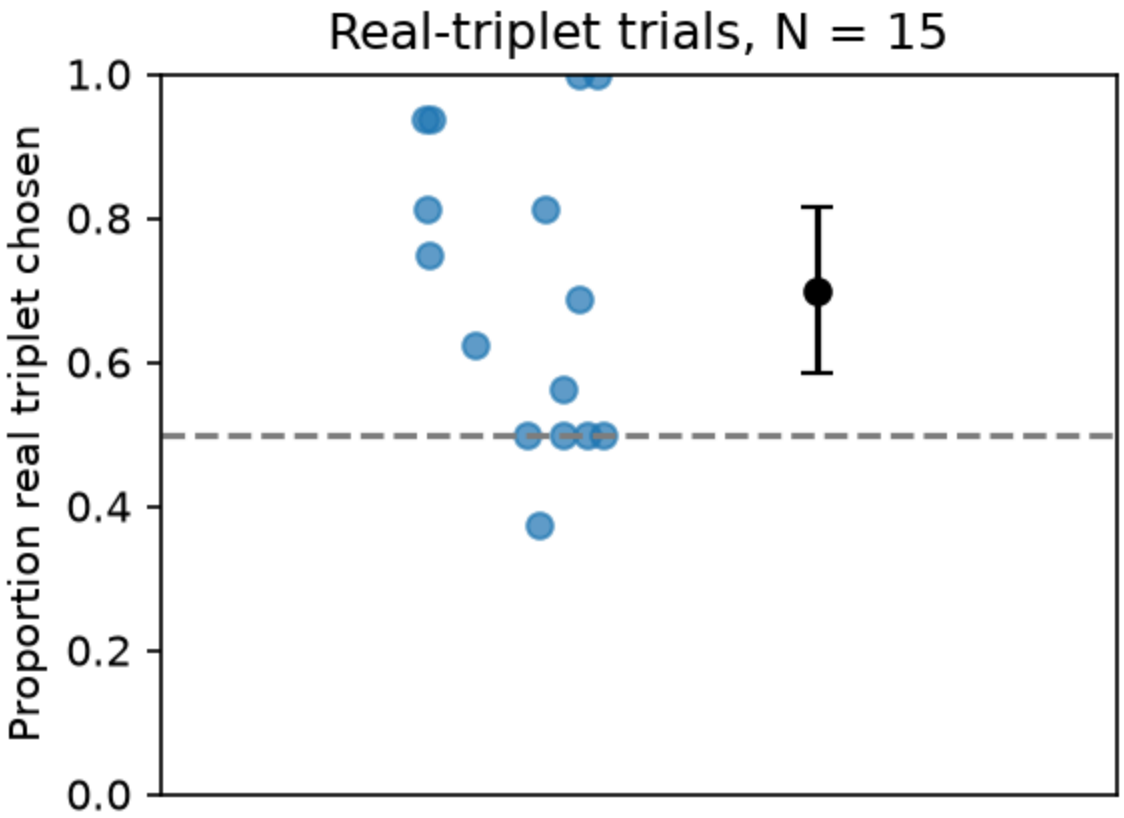
Data and sample. 15 participant files (MATLAB), each with 32 two-alternative familiarity trials: 16 "real" (real colour triplet vs foil) and 16 "pseudo" (pseudo-triplet vs foil). Following the design, pseudo trials are discarded. No participants or trials were excluded (N = 15; 0 excluded): all files were complete, and the recorded `correct` field agreed with the response and the interval order on all 480 trials.

Measure. Per participant, the proportion of the 16 real trials on which the real triplet was chosen.

Primary test (named before analysis). Two-sided one-sample t-test of the 15 proportions against 0.5; effect size d = t/√n, 95% CI by noncentral-t inversion. The claim holds if the mean is above 0.5 with $p < .05$.

Result

	value
Mean proportion correct (real trials)	0.700 (SD 0.209), 95% CI [0.584, 0.816]
t-test vs 0.5	**t(14) = 3.71, p = .0023**
Effect size	**d = 0.96, 95% CI [0.33, 1.56]**
Participants above / at / below 0.5	10 / 4 / 1



Robustness (not primary). Wilcoxon signed-rank vs 0.5: p = .007 (4 ties at 0.5 dropped). Pooled across trials, 168/240 = 0.70 choices of the real triplet (binomial p < .0001; ignores clustering by participant). Descriptively, pseudo-triplet trials were also chosen above chance (mean 0.62, t(14) = 2.36, p = .033); these trials are outside the claim and were not used in the conclusion.

Limits. Small sample (n = 15), so the effect-size interval is wide (d from 0.33 to 1.56); the lower bound is still clearly above zero. The test cannot separate implicit learning from explicit noticing, since participants who reported noticing the triplets were kept by design.

Conclusion. The claim holds on this data: participants chose the real colour triplet over the foil on 70% of real trials, reliably above chance (t(14) = 3.71, p = .002, d = 0.96).